

Teacher Pack: Square and Cube Roots

MYP Mathematics · Grade 8 (MYP Year 3) · Class MYP 8A · 1 lesson × 50 minutes

Files in this package (Lesson 01)

Lesson_01_Preparation.pdf	Student pre-lesson task (10–15 min). Questions P1–P4, T1–T3, N1.
Lesson_01_Class_Worksheet.pdf	In-class worksheet, 8 pages. S1–S6, N1, Examples 1–3, W1–W4, F1–F8, C1–C6, A1–A4, Ch1–Ch2, J1, SM1–SM3, X1–X4 (exit ticket on page 8, printable separately).
Lesson_01_After_Lesson_Practice.pdf	Homework or post-lesson practice. R1, Q1–Q13, ST1–ST3.
Lesson_01_Teacher_Slides.pptx (+ PDF)	22 editable slides with presenter notes.
Answer_Key.pdf	Full worked solutions for every item.
Mini_Whiteboard_Reference.pdf	One-page reference for both mini-whiteboard rounds.

Assumptions made (inputs not specified)

- **Unit/strand:** Number (MYP Criterion A, Knowing and understanding). This is Lesson 01 of the unit.
- **Prior knowledge:** students have met index notation (n^2 , n^3), multiplication facts to 12×12 , area of a square and volume of a cube, and multiplying negative numbers. Likely gaps: confusing n^2 with $2n$, weak recall of squares beyond 10^2 , and sign rules. The Preparation task and the Starter check these.
- **Homework time:** 20–30 minutes (After-Lesson Practice).
- **Calculators:** not used for exact roots or the exit ticket. Used only to *check* estimates.
- **Scope:** principal (positive) square roots only. $x^2 = 25 \Rightarrow x = \pm 5$, surds and simplifying roots are deferred to a later lesson. Square roots of negative numbers are mentioned only as “no real value” (After-Lesson Q10c).
- **Logo:** none supplied, so none is used.

1. Objective, success criteria and alignment

Objective: Calculate and estimate square and cube roots.

Success criteria (worksheet page 1, slide 1). Students can:

- SC1** find the square root of a square number and the cube root of a cube number;
- SC2** explain that a root undoes squaring or cubing, and check by squaring or cubing;
- SC3** state the two consecutive whole numbers a root lies between, and which it is closer to;
- SC4** estimate a root to 1 decimal place and check it.

Alignment table

Objective	Before-lesson preparation	Class modelling and practice	After-lesson questions	Exit-ticket evidence
SC1 exact roots	P1, P2, T1, T2 (square and cube numbers)	Ex 1; W1; MWB R1; F1–F8; C5; A2	Q1–Q4, Q10a–b; ST1	X1, X4
SC2 inverse and checking	P3, P4, worked example, N1(a)	Notice N1; “Check” line in every example; SM1, SM3	Q9, Q10c, Q12	X4 (explain)
SC3 between which integers	N1(b)	Ex 2–3 steps 1–2; W2; MWB R2.1–R2.3; C1, C2, C6, J1	Q5, Q8, Q11; ST2	X2
SC4 estimate to 1 d.p.	(not needed before the lesson)	Ex 2–3 steps 3–4; W2, W3; MWB R2.4–R2.5; C3, C4, A1, SM2	Q6, Q7, Q13; ST3	X3

2. Planning: prerequisites, vocabulary, progression**Prerequisite skills**

- Recall of squares 1^2 – 15^2 and cubes 1^3 – 5^3 , 10^3 (checked by Preparation T1, Starter S1–S6).
- Meaning of index notation: $n^2 = n \times n$, not $2n$ (Preparation P3).
- Multiplying decimals such as 6.3×6.3 (needed for the “trap it” step; calculators may be used for this step if multiplication is slow, because the focus is the reasoning).
- Sign rules: $(-3)^3 = -27$ (Preparation T2c; Remember box).
- Ordering decimals and placing numbers on a number line (J1, C6).

Key vocabulary

square number, cube number, square root ($\sqrt{}$), cube root ($\sqrt[3]{}$), inverse operation, perfect square/perfect cube (used orally as a synonym), estimate, consecutive, approximately equal to ($\approx$), 1 decimal place (1 d.p.).

Difficulty progression

- Exact square roots of small square numbers ($\sqrt{9}$, $\sqrt{25}$) $\rightarrow$ larger ($\sqrt{144}$, $\sqrt{196}$).
- Exact cube roots, including a negative ($\sqrt[3]{-27}$).
- Which two consecutive whole numbers? (bounding)
- Which one is it closer to? (comparing differences)
- Estimate to 1 d.p. by trapping between decimals; check with a calculator.
- Transfer: scaling ($\sqrt{1600}$, $\sqrt{0.16}$), context (area/volume), ordering, reasoning ($\sqrt{a+b} \neq \sqrt{a} + \sqrt{b}$).

Evidence needed to move on

At least about 80% of boards correct in MWB Round 1 (exact roots) before WE DO; at least about 80% correct on R2.1–R2.3 before YOU DO; exit ticket X1–X3 correct for the majority before Lesson 02.

3. Resources

Printed Class Worksheet (1 per student; print page 8 separately if you want to collect exit tickets), mini-whiteboards, pens and erasers, scientific calculators (for checking only, from 10:00), slides, visualiser or board. Printed After-Lesson Practice for homework. Collect Preparation sheets at the door or look at them during the Starter.

4. Minute-by-minute lesson plan (50 minutes)

Time	Stage	Teacher action	Student action	Resource	Evidence / decision
00:00–03:00	Starter (3)	Greet at door; students start immediately. Glance at Preparation sheets while circulating.	Complete S1–S6 silently.	WS p1; slide 2	Note who is stuck on 11^2 , 10^3 or confuses 3^3 with 9.
03:00–04:00	Starter feedback (1)	Reveal answers; students self-mark with a tick or correction.	Self-mark; hands up for scores.	Slide 3	Fewer than 4/6 correct: use the Support Box table later.
04:00–07:00	Prerequisite check / Preparation link (3)	Ask N1 from Preparation: square area 49 and area 50. Address “ $5^2 = 10$ ” error (P3). Students without Preparation: they answer from the Starter.	Answer on boards: side for area 49; “between what” for area 50.	Slide 4	If many say $5^2 = 10$, do 30-second reteach with the dot-square picture.
07:00–08:00	I DO: concept (1)	Inverse operation diagram: side $\rightarrow$ area and back. Define $\sqrt{\quad}$, $\sqrt[3]{\quad}$.	Listen; complete Notice N1.	Slide 5; WS p2 N1	Cold-call: “What undoes squaring?”
08:00–10:00	I DO Example 1 (2)	Model $\sqrt{81}$, $\sqrt[3]{125}$, $\sqrt[3]{-27}$ with a check each time.	Predict first, then annotate.	Slides 6–7; WS p2	Prediction of $\sqrt[3]{-27}$ shows sign understanding.
10:00–13:00	I DO Example 2 (3)	Model estimate of $\sqrt{40}$: 4 steps plus number line. Calculators out for the check.	Annotate each step in “My notes”.	Slides 8–9; WS p2	Ask “closer to 6 or 7? Why?” before revealing.
13:00–15:00	I DO Example 3 (2)	Model $\sqrt[3]{50}$ with the same 4 steps; stress “cube, not square”.	Annotate.	Slide 10; WS p3	Watch for squaring instead of cubing.
15:00–19:00	Mini-whiteboard Round 1 (4)	5 questions, routine each time. Scan and record.	Answer on boards.	Slides 11–12; MWB Reference	Threshold: about 80% correct. Below: 2–3 min reteach (see Section 7), taken from YOU DO time.
19:00–21:00	WE DO W1 (2)	Heavily supported: fill in blanks together.	Fill blanks on WS p3.	Slide 13; WS p3	Choral response.
21:00–24:00	WE DO W2 (3)	Partly supported: students do each step, then compare with a partner.	Complete blanks. W2	Slide 13	Check “closer to 8” reasoning.

Time	Stage	Teacher action	Student action	Resource	Evidence / decision
24:00–26:00	WE DO W3 (2)	First step only given. Students try, then teacher reveals.	Attempt W3 alone.	Slide 14	Scan books: cubing correctly?
26:00–27:00	WE DO W4 (1)	No support: $\sqrt{1600}$, $\sqrt{0.16}$.	Attempt, then check.	Slides 14–15	Common error $\sqrt{0.16} = 0.04$: address on slide 15.
27:00–31:00	Mini-whiteboard Round 2 (4)	5 estimation questions, routine.	Answer on boards.	Slides 16–17	About 80% on R2.1–R2.3 before YOU DO; if not, reteach the “closer to” step (Section 7).
31:00–41:00	YOU DO (10)	30-second transition. Silent 5 min, then circulate: Foundation first for students flagged earlier. Mini-plenary at 38:00 on C3 if needed.	Foundation → Core → J1 → Apply → Challenge.	Slide 18; WS pp4–6	Target: all of F and C1–C4. Unfinished Core becomes the Lesson 02 starter.
41:00–44:00	Check for understanding: Spot the Mistake (3)	Show SM1–SM3. Think–pair–share on SM2. Reveal corrections.	Identify first error, explain, correct (WS pp6–7).	Slides 19–20	SM2 (“middle” error) shows understanding of “closer to”.
44:00–49:00	Exit Ticket (5)	Silent, no calculators. Collect.	Complete X1–X4.	Slide 21; WS p8	Sort into 3 piles tonight (Section 10).
49:00–50:00	Recap and next step (1)	Recap 4-step method; set After-Lesson Practice.	Rate SC1–SC4 with fingers.	Slide 22	Homework due next lesson.
Total: 3+1+3+1+2+3+2+4+2+3+2+1+4+10+3+5+1 = 50 minutes					

5. Preparation follow-up

- **During the Starter (00:00–03:00):** walk the room, look at T1 (table) and P3 (Omar $5^2 = 10$). Note names who ticked “I need help”.
- **Two-minute response plan** (built into slide 4): (1) If the $5^2 = 10$ error is common, draw a 5×5 dot square and count 25. (2) If $(-4)^2$ answered as -16 : “negative times negative is positive”. (3) N1(b): collect predictions for the square with area 50, and leave them on the board to return to in Example 2.
- **Students who did not complete Preparation:** they are not excluded. The Starter S1–S6 covers the same facts, and the Support Box (WS p4) gives the square and cube table. Seat them near the front for I DO. Ask them to complete the Preparation as part of the homework.

6. Board plan

Board (left to right)

Title: Square and Cube Roots

Objective: Calculate and estimate square and cube roots.

Vocabulary: square root $\sqrt{\quad}$
cube root $\sqrt[3]{\quad}$
inverse, estimate, consecutive,
 $\approx$

Remember:

$$6^2 = 36 \Leftrightarrow \sqrt{36} = 6$$

$$4^3 = 64 \Leftrightarrow \sqrt[3]{64} = 4$$

$\sqrt{\quad}$ means the positive root.

Method: estimate a root

1. Square/cube numbers either side
2. Closer to? (compare differences)
3. Trap it with decimals
4. Choose the closer one; check

Worked examples:

$$\sqrt{81} = 9 \text{ since } 9^2 = 81$$

$$\sqrt[3]{-27} = -3 \text{ since } (-3)^3 = -27$$

$$36 < 40 < 49 \Rightarrow 6 < \sqrt{40} < 7$$

$$6.3^2 = 39.69, 6.4^2 = 40.96$$

$$\sqrt{40} \approx 6.3$$

Common mistake:

$$\sqrt{16} \neq 8 \text{ (not } \div 2)$$

$$\sqrt{60} \neq 7.5 \text{ (not the middle)}$$

Your Turn:

WS pp4–5: F1–F8, C1–C6

then J1, then A1–A4

Challenge: Ch1–Ch2

7. Teaching script

SAY = simple, precise words. WRITE = exact board text. ASK = questions to pose. EXPECT = likely correct responses. WATCH FOR = errors. IF STUDENTS STRUGGLE = the next move.

TP1 Starter and prerequisite check (00:00–07:00, slides 2–4)

SAY	"Start the six starter questions on page 1 in silence. You have three minutes. No calculators."
WRITE	Starter: S1–S6, page 1. 3 minutes.
ASK	"What does the small 2 in 8^2 mean?" "In the Preparation, a square had area 49. How long was its side? What about area 50?"
EXPECT	S1 64, S2 121, S3 27, S4 1000, S5 64, S6 25, 36, 81. Side 7 cm; area 50 gives a side "a bit more than 7".
WATCH FOR	$3^3 = 9$; $8^2 = 16$; circling 20 or 48 as square numbers; " $\sqrt{50} = 25$ ".
IF STUDENTS STRUGGLE	Draw a 3×3 grid and count. Point to the Support Box table (WS p4) as a reference for the rest of the lesson.

TP2 Concept: the root undoes the power (07:00–08:00, slide 5)

SAY	“Squaring takes the side of a square and gives the area. A square root goes backwards: it takes the area and gives the side. So $\sqrt{36} = 6$ because $6^2 = 36$. The square root sign means the positive root. A cube root undoes cubing: $\sqrt[3]{64} = 4$ because $4^3 = 64$.”
WRITE	$6^2 = 36 \Leftrightarrow \sqrt{36} = 6$ $4^3 = 64 \Leftrightarrow \sqrt[3]{64} = 4$
ASK	“What operation undoes squaring?” “What is the side of a square with area 25?”
EXPECT	“Square root.” “5 cm, because $5 \times 5 = 25$.”
WATCH FOR	“Divide by 2” (confusing square with double).
IF STUDENTS STRUGGLE	Use the area pictures on WS p2 (N1): count along one side of the 5×5 square.

TP3 I DO Example 1: exact roots (08:00–10:00, slides 6–7)

SAY	“To find $\sqrt{81}$ I ask: which number times <i>itself</i> gives 81? $9 \times 9 = 81$, so $\sqrt{81} = 9$. I always check: $9^2 = 81$. For $\sqrt[3]{125}$ I ask: which number times itself times itself gives 125? $5 \times 5 = 25$, $25 \times 5 = 125$. For $\sqrt[3]{-27}$: a negative times a negative is positive, times another negative is negative again. So $(-3)^3 = -27$, and $\sqrt[3]{-27} = -3$.”
WRITE	$\sqrt{81} = 9$ check $9^2 = 81$ ✓ $\sqrt[3]{125} = 5$ check $5^3 = 125$ ✓ $\sqrt[3]{-27} = -3$ check $(-3)^3 = -27$ ✓
ASK	“Predict on your fingers: is $\sqrt[3]{-27}$ positive, negative, or impossible?”
EXPECT	Mixed predictions; after modelling: negative.
WATCH FOR	$\sqrt[3]{125} = 25$ (only squared); $\sqrt[3]{-27}$ “impossible”.
IF STUDENTS STRUGGLE	Say the check aloud as a sentence: “5 cubed is 5 times 5 times 5.”

TP4 I DO Example 2: estimate $\sqrt{40}$ (10:00–13:00, slides 8–9)

SAY	“40 is not a square number, so $\sqrt{40}$ is not a whole number. Step 1: find the square numbers either side, 36 and 49. So $\sqrt{40}$ is between 6 and 7. Step 2: 40 is 4 away from 36 and 9 away from 49, so it is closer to 6. Step 3: I test decimals. $6.3^2 = 39.69$, too small. $6.4^2 = 40.96$, too big. Step 4: 39.69 is only 0.31 away from 40, and 40.96 is 0.96 away. So $\sqrt{40} \approx 6.3$. Now check with a calculator: 6.324...”
WRITE	$36 < 40 < 49$ $6 < \sqrt{40} < 7$, closer to 6 $6.3^2 = 39.69$, $6.4^2 = 40.96$ $\sqrt{40} \approx 6.3$ (1 d.p.)
ASK	“Why did I start testing at 6.3 and not 6.8?” “What does $\approx$ mean?”
EXPECT	“Because 40 is closer to 36, so the root is closer to 6.” “Approximately equal to.”
WATCH FOR	Taking the halfway point (6.5); squaring 6.3 as 6.3×2 ; writing = instead of $\approx$.
IF STUDENTS STRUGGLE	Use a calculator for the decimal squaring. The key reasoning is bounding and comparing, not long multiplication.

TP5 I DO Example 3: estimate $\sqrt[3]{50}$ (13:00–15:00, slide 10)

SAY	“Same four steps, but now with <i>cube</i> numbers. 27 and 64 are either side of 50, so $\sqrt[3]{50}$ is between 3 and 4. 50 is 14 from 64 and 23 from 27, so closer to 4. $3.7^3 = 50.653$ is just above, $3.6^3 = 46.656$ is below. So $\sqrt[3]{50} \approx 3.7$.”
WRITE	$27 < 50 < 64 \Rightarrow 3 < \sqrt[3]{50} < 4$ $3.6^3 = 46.656$, $3.7^3 = 50.653$ $\sqrt[3]{50} \approx 3.7$
ASK	“Which cube numbers are either side of 50? How do you know 49 is not one of them?”
EXPECT	“27 and 64. 49 is a square number, not a cube number.”
WATCH FOR	Using square numbers (49, 64) for a cube root.
IF STUDENTS STRUGGLE	Point to the cube row of the Support Box. Underline the small 3 on the root sign.

TP6 Mini-whiteboard Round 1: exact roots (15:00–19:00, slides 11–12)

Routine (every question): “Boards down → work independently → show in 3, 2, 1 → do not erase → scan all boards.”

#	Question	Answer	Misconception / check	If many miss it
R1.1	$\sqrt{9}$ (very easy)	3	Answer 4.5: halving.	Show 3×3 grid; “times <i>itself</i> ”.
R1.2	$\sqrt{100}$ (easy)	10	Answer 50.	Ask “What times itself is 100?”
R1.3	$\sqrt[3]{64}$ (similar)	4	Answer 8: using square root.	Point to the small 3: $4 \times 4 \times 4$.
R1.4	A square has area 81 cm^2 . Side? (variation)	9 cm	Units missing or 40.5.	Link to N1 area diagram.
R1.5	True or false: $\sqrt{36} = 18$ (misconception)	False; $\sqrt{36} = 6$	“True” shows the halving belief.	Check: $18^2 = 324$, not 36. Every root answer must be checked.

Decision: about 80% or more correct on R1.1–R1.5 → go to WE DO. Below about 80% → **2–3 minute reteach:** re-do Example 1 with $\sqrt{49}$ and $\sqrt[3]{8}$, saying the check aloud; then **re-check** with $\sqrt{121}$ and $\sqrt[3]{27}$ on boards. Take the time from YOU DO (reduce target to F1–F8 and C1–C3).

TP7 WE DO W1–W4 with fading support (19:00–27:00, slides 13–15)

W1 (full support)	Prompt: “Fill the blanks: what number times itself is 144?” Expect: “12.” $12 \times 12 = 144$ so $\sqrt{144} = 12$; check $12^2 = 144$. $10 \times 10 \times 10 = 1000$, so $\sqrt[3]{1000} = 10$. Watch for: $\sqrt{144} = 72$.
W2 (partial)	Prompt: “Which square numbers are either side of 70? Which is 70 closer to?” Expect: $64 < 70 < 81$, so $8 < \sqrt{70} < 9$; $70 - 64 = 6$, $81 - 70 = 11$, closer to 8. Try $8.3^2 = 68.89$, $8.4^2 = 70.56$. Differences 1.11 and 0.56, so $\sqrt{70} \approx 8.4$. Calculator 8.366... ✓. Watch for: stopping at 8.3 because “closer to 8”. Closer to 8 does not mean 8.1 or 8.2; the trapping step decides.
W3 (first step only)	Prompt: “You have the first step. Finish it on your own in 90 seconds.” Expect: $8 < 20 < 27$ so $2 < \sqrt[3]{20} < 3$; $20 - 8 = 12$, $27 - 20 = 7$, closer to 3. $2.7^3 = 19.683$, $2.8^3 = 21.952$. Differences 0.317 and 1.952, so $\sqrt[3]{20} \approx 2.7$. Calculator 2.714... ✓. Watch for: squaring 2.7.
W4 (no support)	Prompt: “Use $\sqrt{16} = 4$.” Expect: $40^2 = 1600$, so $\sqrt{1600} = 40$. $0.4^2 = 0.16$, so $\sqrt{0.16} = 0.4$. Watch for: $\sqrt{0.16} = 0.04$. Check: $0.04^2 = 0.0016$.
IF STUDENTS STRUGGLE	Return to the Support Box sentence stem. Pair a secure student with a less secure one for W2 only.

TP8 Mini-whiteboard Round 2: estimating (27:00–31:00, slides 16–17)

#	Question	Answer	Misconception / check	If many miss it
R2.1	$\sqrt{20}$ is between which two whole numbers? (very easy)	4 and 5	“10 and 11” (halving).	List squares: 16, 25.
R2.2	$\sqrt{45}$: between? closer to? (easy)	6 and 7; closer to 7	“closer to 6” (smaller number first).	Differences: $45 - 36 = 9$, $49 - 45 = 4$.
R2.3	$\sqrt[3]{30}$: between? (similar)	3 and 4	“5 and 6” (used squares).	Cube numbers 27 and 64.
R2.4	Estimate $\sqrt{10}$ to 1 d.p. (variation)	3.2	3.1 chosen without checking; 5 (halving).	$3.1^2 = 9.61$, $3.2^2 = 10.24$; differences 0.39 and 0.24.
R2.5	True or false: “ $\sqrt{65} \approx 8.5$ because it is between 8 and 9” (misconception)	False; ≈ 8.1	“True” shows the halfway belief.	65 is only 1 above 64, so the root is just above 8. $8.1^2 = 65.61$.

Decision: about 80% correct on R2.1–R2.3 is required to start YOU DO. If fewer, do a **3-minute reteach** of Steps 1–2 only with $\sqrt{27}$ (between 5 and 6, closer to 5), then **re-check** with $\sqrt{50}$ and $\sqrt[3]{10}$ on boards. If R2.4–R2.5 are weak but R2.1–R2.3 are secure, start YOU DO and pull a small group for C3 at 34:00.

TP9 YOU DO (31:00–41:00, slide 18)

SAY	“Page 4. Foundation, then Core, then J1, then Apply. Challenge only if you finish. Five minutes silent, then I will come round. Check every answer by squaring or cubing.”
WRITE	YOU DO: F1–F8 → C1–C6 → J1 → A1–A4 → Ch1–Ch2. Finish time: 41:00.
ASK	(circulating) “Show me your check for C3.” “Why is $\sqrt{45}$ smaller than 7?”
EXPECT	Most complete F and C1–C4 in 10 minutes; confident students reach A1–A3.
WATCH FOR	C6 ordering: treating $\sqrt[3]{343}$ as $\sqrt{343}$. A3: $\sqrt{0.09} = 0.03$. A4: choosing 42 ($6.5^2 = 42.25$, so 42 is closer to 6).
IF STUDENTS STRUGGLE	Mini-plenary at 38:00 on the most common error. Students stuck on F: use the Support Box table and the question “what times itself?”

TP10 Spot the Mistake (41:00–44:00, slides 19–20)

SAY	“Each student made one first mistake. Find it, explain it, and correct it.”
ASK	“What did Bea assume that isn’t true?”
EXPECT	SM1: halved instead of finding a number times itself; $\sqrt{16} = 4$. SM2: assumed the root is in the middle; 60 is closer to 64, so $\sqrt{60} \approx 7.7$. SM3: $(-2)^3 = -8$, so $\sqrt[3]{-8} = -2$.
WATCH FOR	Students who “correct” SM2 to 7.8 without checking ($7.7^2 = 59.29$ is closer).
IF STUDENTS STRUGGLE	Ask them to square Bea’s answer: $7.5^2 = 56.25$, which is not close to 60.

TP11 Exit ticket and recap (44:00–50:00, slides 21–22)

SAY	“Exit ticket, page 8, on your own, no calculator. Five minutes.” Then: “Today: roots undo powers; estimate in four steps; always check.”
WRITE	Homework: After-Lesson Practice (20–30 min). Required R1, Q1–Q10, Self-Test.
ASK	“Show fingers 1–4 for how confident you are with SC4.”
EXPECT	X1 7, 3; X2 9 and 10; X3 9.2; X4 20 cm because $20^3 = 8000$.
WATCH FOR	X3 answer 9.5 (halfway); X4 answer 2666.7 (divided by 3).
IF STUDENTS STRUGGLE	Collect anyway; use the sorting in Section 10.

8. Misconceptions

Misconception	Example	Diagnosis	Teacher response
Square root means halve	$\sqrt{16} = 8$	Confuses “square” with “double”; weak meaning of n^2 .	Draw a 4×4 grid: area 16, side 4. “What times <i>itself</i> ?” Always check $8^2 = 64 \neq 16$.
Cube root means divide by 3	$\sqrt[3]{27} = 9$	Confuses cubing with tripling.	$9^3 = 729$. Build a $3 \times 3 \times 3$ cube picture.
The root is halfway between the two whole numbers	$\sqrt{60} = 7.5$	Assumes the scale is linear; does not compare distances.	Compare $60 - 49 = 11$ with $64 - 60 = 4$. Check $7.5^2 = 56.25$.
Negative numbers have no cube root	“ $\sqrt[3]{-8}$ has no answer”	Over-generalises from square roots.	$(-2)^3 = (-2)(-2)(-2) = -8$. Contrast with $\sqrt{-16}$: no real number squares to a negative.
Root of a decimal: count decimal places wrongly	$\sqrt{0.16} = 0.04$	Takes the root of the digits only.	Check: $0.04^2 = 0.0016$. Use $0.4 \times 0.4 = 0.16$.
Roots split over addition	$\sqrt{9 + 16} = 3 + 4 = 7$	Over-generalises the distributive law.	Work out the inside first: $\sqrt{25} = 5$. Check $7^2 = 49$.
Uses squares for cube roots	$\sqrt[3]{50}$ between 7 and 8	Ignores the index on the root sign.	Circle the small 3; use the cube row of the Support Box.

9. Differentiation

- **Support:** Support Box (square and cube table, method, sentence stem); W1–W2 blanks; allow calculators for the decimal-squaring step only; Foundation questions start from $\sqrt{25}$; seat students who missed Preparation near the front.
- **Stretch:** A4 (range of n), Ch1 (square-and-cube numbers: 64, 729), Ch2 ($\sqrt{a+b} \neq \sqrt{a} + \sqrt{b}$), homework Q11–Q13. Ask “Can you find a number whose square root is 7.5 exactly?” (56.25).
- **Language support:** vocabulary table on WS p1; sentence stem in the Support Box; SAY lines use short sentences.

10. Exit ticket interpretation and next-lesson decisions

Essential objective = X1 (exact roots) and X2–X3 (estimate). X4 is transfer.

Most students secure (X1–X3 correct for about 80% or more)	Lesson 02 starts with a 3-question retrieval, then moves on (e.g. solving $x^2 = k$ giving $\pm$, or roots in Pythagoras). Use A4 and Ch1 as a do-now for early finishers.
About half struggling	8-minute guided group at the start of Lesson 02 using W2-style scaffolds while others do the unfinished Core and Apply items. Re-teach Step 2 (“closer to”) with a number line. Re-check with a 2-question board round.
Most struggling	Revisit prerequisites: squares and cubes recall (Preparation T1) and the inverse idea (N1 area pictures). Re-teach Example 1 and Example 2 with new numbers ($\sqrt{121}$, $\sqrt{30}$), then repeat MWB Round 2 before any new content.
X1 wrong specifically	Revisit meaning of n^2 and n^3 (halving/tripling misconception).
X2 right but X3 wrong	Revisit decimal squaring and Step 4 (compare differences); allow calculator for the squaring step.
X4 wrong only	Transfer issue: link volume = edge ³ and scaling $\sqrt[3]{8000} = \sqrt[3]{8} \times 10$.

Homework follow-up: release the After-Lesson Practice answers (Answer Key, section “After-Lesson Practice”) after marking. Look at Q6 and Q8 for the same errors as X3 and X2.